

Architecture

[Services List](#)

[Services Diagram](#)

[Used design patterns, standards & conventions](#)

Security & Cryptography

[HTTP Layer](#)

[Interservice Layer](#)

[Database Layer](#)

PII

Codebase

[Git Repositories list](#)

<https://gitlab.com/tbrain-ru/aid/aid-core>

<https://gitlab.com/tbrain-ru/aid/transactions-monitoring-service>

<https://gitlab.com/tbrain-ru/aid/stat-service>

<https://gitlab.com/tbrain-ru/aid/submissions-service>

<https://gitlab.com/tbrain-ru/aid/checkups-service>

<https://gitlab.com/tbrain-ru/aid/aid-scoring-service>

<https://gitlab.com/tbrain-ru/aid/audit-service>

https://gitlab.com/tbrain-ru/aid/documents_service

<https://gitlab.com/tbrain-ru/aid/user-behaviour-tracking-service>

<https://gitlab.com/tbrain-ru/aid/passport-service>

<https://gitlab.com/tbrain-ru/aid/passport-data-service>

<https://gitlab.com/tbrain-ru/aid/auth-service>

<https://gitlab.com/tbrain-ru/services/tfa-service>

<https://gitlab.com/tbrain-ru/services/notification-service>

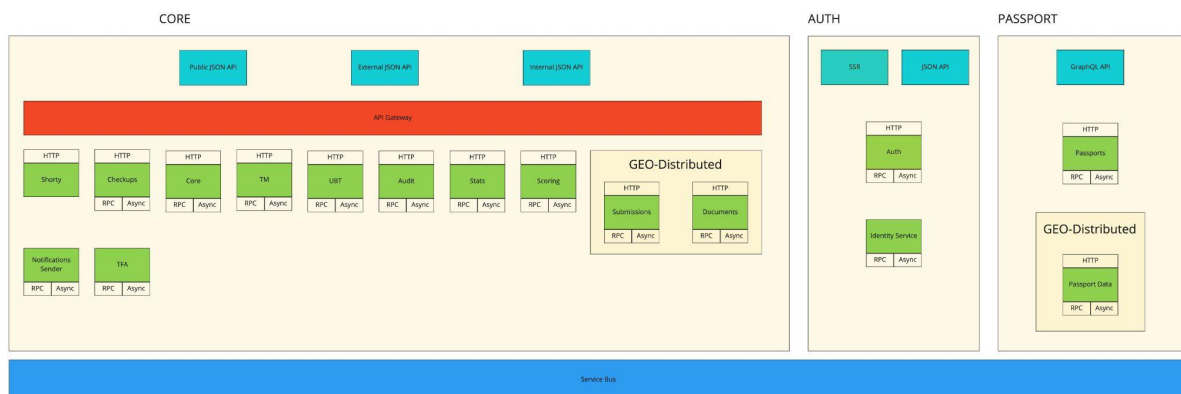
[Code Listings Examples](#)

Architecture

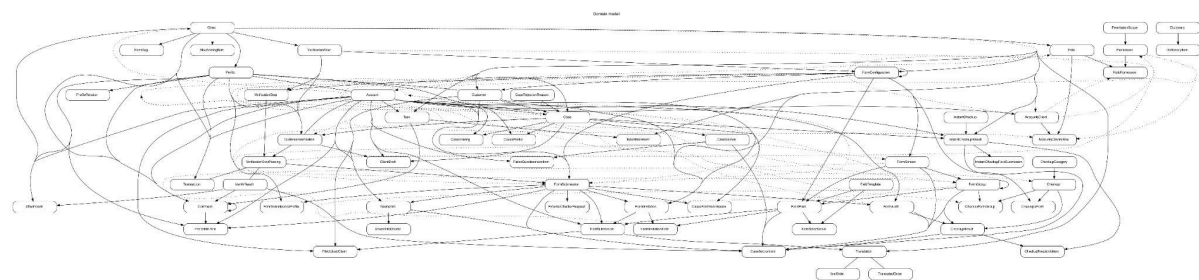
Services List

1. **Core.** Provides multi-tenancy, billing and some compliance tools.
2. **Transactions Monitoring (TM).** Toolset for monitoring transaction & working with risks.
3. **Submissions.** Geo-distributed service for storing customers submissions data (including PII).
4. **Statistics.** Collection, aggregation and presentation of analytical data.
5. **Checkups.** Performing AML, KYC, KYB etc checkups.
6. **Scoring.** Performs calculations of risk scores.
7. **Documents.** Geo-distributed service for storing user-uploaded documents.
8. **User Behavior Tracking.** Service for collecting user's interactions during form filling.
9. **Audit.** Tracking actions of Compliance Officers and other users.
10. **Passport.** Core of A.ID Passport service.
11. **Passport Data.** Geo-distributed service for storing A.ID Passports.
12. **Auth.** SSO OAuth2.0 solution.
13. **2FA.** Sends and verifies 2FA codes.
14. **Notifications.** Sends SMSs, Emails, and performs voice calls to customers.
15. **Shorty.** Links shortifier.

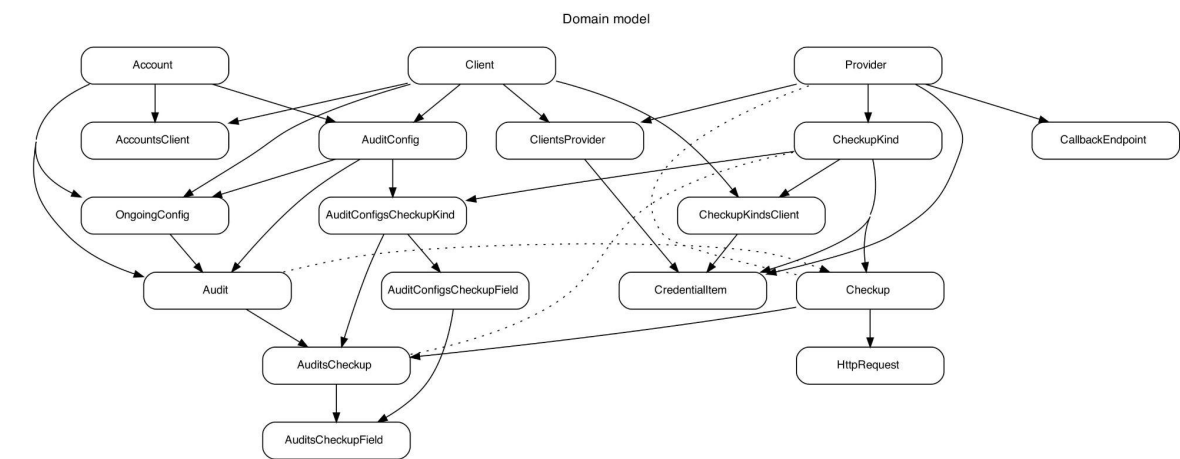
Services Diagram



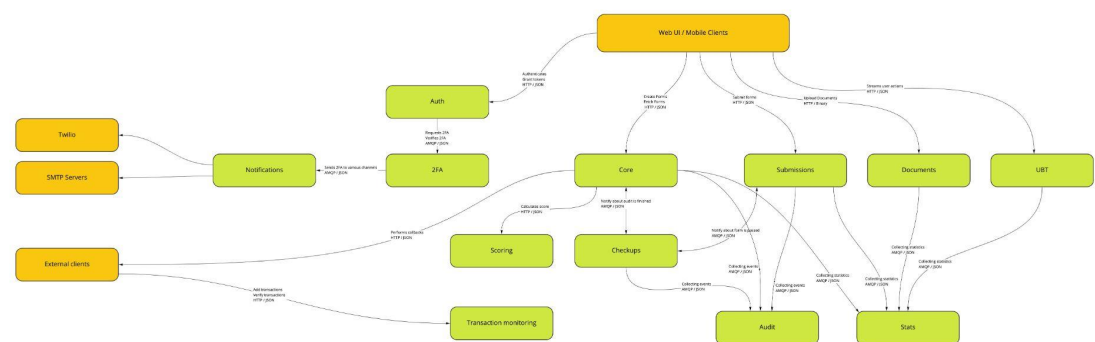
ER (Core)



ER (Checkups)



DFD (Common scenario)



Used design patterns, standards & conventions

1. **Service Bus** (<https://www.enterpriseintegrationpatterns.com/MessageBus.html>) - for interservice communication
2. **MVC** (<https://en.wikipedia.org/wiki/Model%E2%80%93view%E2%80%93controller>) - Web Apps base structure
3. **Choreography**
(<https://docs.microsoft.com/en-us/azure/architecture/patterns/choreography>) - for distributed tasks and transactions
4. **Sidecar** (<https://docs.microsoft.com/en-us/azure/architecture/patterns/sidecar>) - for logs extraction
5. **CQRS** (<https://docs.microsoft.com/en-us/azure/architecture/patterns/cqrs>) - for maximizing performance and scalability
6. **Event Sourcing**
(<https://docs.microsoft.com/en-us/azure/architecture/patterns/event-sourcing>) - for user actions audit
7. **Geode and Geo-Distributed Services**
(<https://docs.microsoft.com/en-us/azure/architecture/patterns/geodes>) - for storing PII in customer's country
8. **Pub-Sub**
(<https://docs.microsoft.com/en-us/azure/architecture/patterns/publisher-subscriber>) - for interservice async communication

Security & Cryptography

HTTP Layer

- SSL
- CORS

Interservice Layer

- Encryption of all the RabbitMQ messages with AES256-CBC

Database Layer

- Encryption of all the PII data with AES256-CFB

PII

We use following approaches for dealing with PII data:

1. Geo-Distribution. All the sensitive user data is stored only in the user's region.
2. Cryptography. We encrypt all the sensitive data in DB.
3. Blind Indexes for search and filtering across the data.

Codebase

Git Repositories list

Ruby

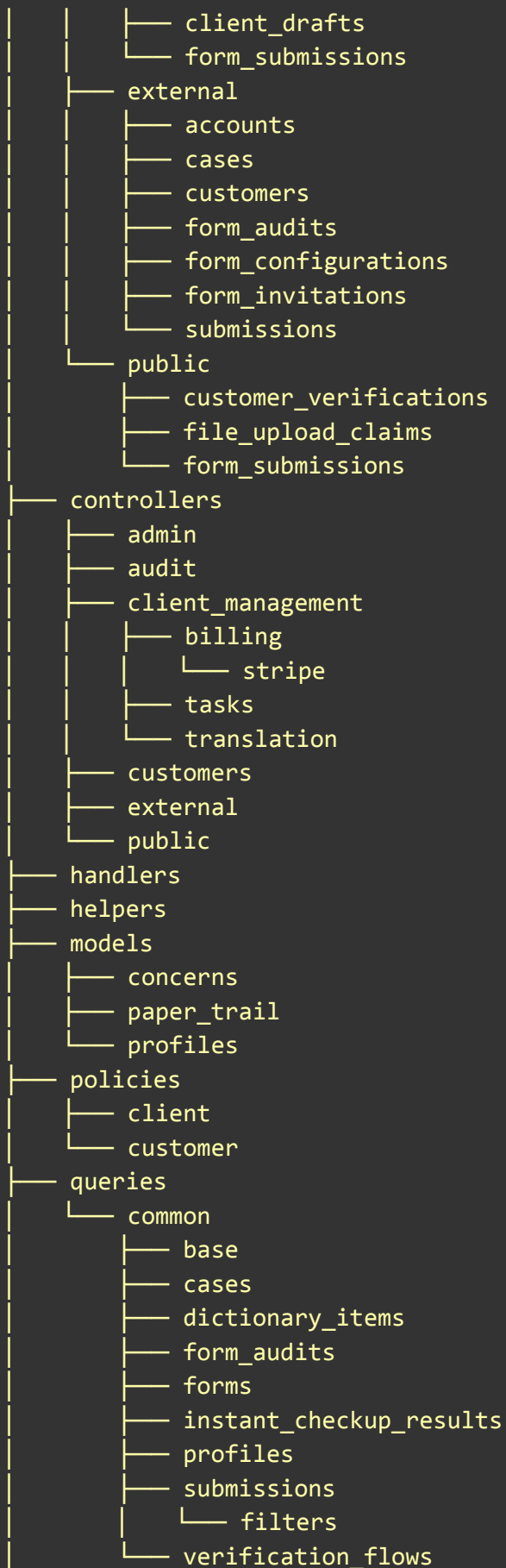
1. <https://gitlab.com/tbrain-ru/aid/aid-core>

- a. **Description:** Repo for Core Service.
- b. **Structure:**



- instant_checkup_results
- internal_onboarding
- profile_info_items
- profiles
- risk_scoring_items
- roles
- submissions
- task_attachments
- task_comments
- tasks
- translated
- verification_flows
- common
 - accounts
 - audits
 - billing
 - callbacks
 - cases
 - clients
 - counter_statistics
 - customer_verifications
 - customers
 - device_infos
 - execute_with_billing
 - instant_checkups
 - submissions
 - translations
 - verification_step_passings
 - finish
 - start
 - form_invitations
 - forms
 - idenfy_results
 - instant_checkups
 - integrations
 - idenfy
 - submissions
 - misc
 - verification_step_passings
- customer
 - client_drafts
 - internal_personal_onboarding
 - submissions
- events
- external
 - cases

- forms
- submissions
- misc
- permissions
- public
 - file_upload_claims
 - idenfy_results
 - submissions
 - verifications
- contracts
 - admin
 - accounts
 - case_rejection_reasons
 - checkup_categories
 - checkups
 - clients
 - field_templates
 - instant_checkups
 - regions
 - audit
 - instant_checkups
 - client
 - accounts_clients
 - billing
 - bulk
 - case
 - case_sections
 - case_sharings
 - checkups
 - comments
 - customer_verifications
 - customers
 - form_configuration
 - form_invitations
 - form_tags
 - profile_info_items
 - profiles
 - risk_scoring_items
 - roles
 - task_attachments
 - task_comments
 - tasks
 - verification_flows
- common
 - accounts
- customer




```
|— serializers
|   |— admin
|   |— audit
|   |— client
|   |   |— case_divided_search
|   |   |— export
|   |   |   |— cases
|   |   |   |— instant_checkups
|   |   |— translations
|   |— common
|   |— customer
|   |— external
|   |— public
|— services
|   |— audit_service
|   |— billing
|   |— checkups
|   |— comply_advantage
|   |— core
|   |— documents
|   |— fingerprintjs
|   |— ibm_translate
|   |— idenfy
|   |— intercom
|   |— notifications
|   |— passport_data
|   |— pdf
|   |— rabbit
|   |— reports
|   |— shorty
|   |— sso
|   |— stubs
|   |— submissions
|   |— tfa
|   |— translated
|— uploaders
|— validators
|— workers
|— bin
|— config
|— db
|   |— migrate
|   |— seeds
|   |   |— dev
|— lib
|— auth
```

```

├── strategies
├── caching
├── sidekiq
├── middleware
├── tasks
├── spec
├── commands
├── factories
├── fixtures
├── requests
├── external
├── shared
├── auth
├── rpc_clients
├── support
├── system
├── boot

```

c. Dependencies:

```

* aasm (5.1.1)
* activemodel (6.0.4.1)
* activerecord (6.0.4.1)
* activesupport (6.0.4.1)
* addressable (2.8.0)
* amq-protocol (2.3.2)
* authy (3.0.0)
* aws-eventstream (1.1.0)
* aws-partitions (1.416.0)
* aws-sdk-core (3.111.0)
* aws-sdk-kms (1.41.0)
* aws-sdk-s3 (1.87.0)
* aws-sigv4 (1.2.2)
* bcrypt (3.1.16)
* blueprinter (0.25.2)
* bond (0.5.1)
* bundler (2.2.15)
* bunny (2.15.0)
* bunny_encrypted (0.1.0 fd45d2f)
* byebug (11.1.3)
* carrierwave (2.2.0)
* carrot-top (0.0.7)
* coderay (1.1.3)
* concurrent-ruby (1.1.9)
* connection_pool (2.2.3)
* counter_culture (2.7.0)

```

- * database_cleaner-active_record (2.0.1)
- * database_cleaner-core (2.0.1)
- * database_cleaner-redis (2.0.0)
- * declarative (0.0.20)
- * declarative-builder (0.1.0)
- * declarative-option (0.1.0)
- * diff-lcs (1.4.4)
- * disposable (0.4.7)
- * domain_name (0.5.20190701)
- * dotenv (2.7.6)
- * dry-auto_inject (0.7.0)
- * dry-configurable (0.12.0)
- * dry-container (0.7.2)
- * dry-core (0.5.0)
- * dry-equalizer (0.3.0)
- * dry-events (0.2.0)
- * dry-inflector (0.2.0)
- * dry-initializer (3.0.4)
- * dry-logic (1.1.0)
- * dry-monitor (0.3.2)
- * dry-schema (1.5.6)
- * dry-struct (1.3.0)
- * dry-system (0.18.1)
- * dry-types (1.4.0)
- * dry-validation (1.6.0)
- * e2mmap (0.1.0)
- * et-orbi (1.2.4)
- * factory_bot (6.2.0)
- * faker (2.18.0)
- * faraday (1.3.0)
- * faraday-net_http (1.0.1)
- * ffi (1.15.0)
- * fugit (1.4.2)
- * hashie (4.1.0)
- * hirb (0.7.3)
- * http-accept (1.7.0)
- * http-cookie (1.0.3)
- * httpclient (2.8.3)
- * hutch (1.0.0)
- * i18n (1.8.10)
- * ice_nine (0.11.2)
- * image_processing (1.12.1)
- * jmespath (1.4.0)
- * json (2.5.1)
- * jwt (2.2.2)
- * jwt_sessions (2.5.2)

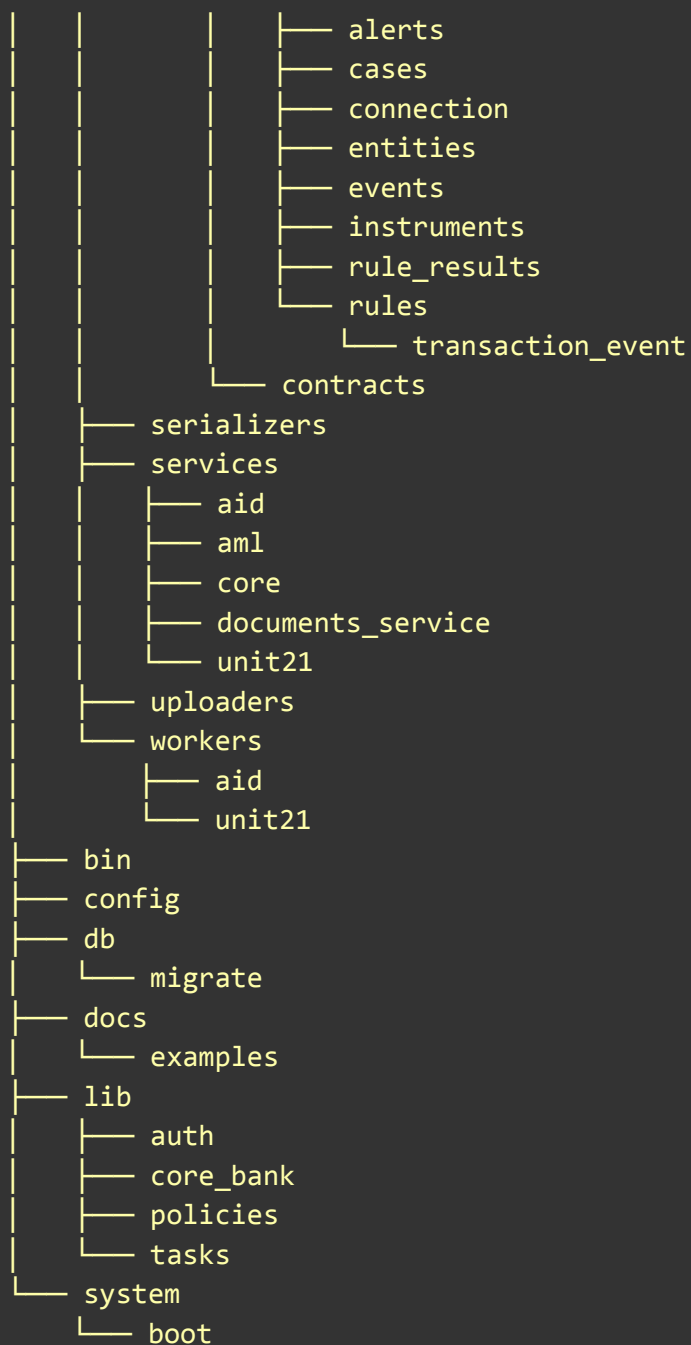
- * launchy (2.5.0)
- * letter_opener (1.7.0)
- * mail (2.7.1)
- * method_source (1.0.0)
- * mime-types (3.3.1)
- * mime-types-data (3.2021.0225)
- * mimemagic (0.3.8)
- * mini_magick (4.11.0)
- * mini_mime (1.0.2)
- * mini_portile2 (2.6.1)
- * minitest (5.14.4)
- * multi_json (1.15.0)
- * multi_xml (0.6.0)
- * multipart-post (2.1.1)
- * mustermann (1.1.1)
- * netrc (0.11.0)
- * nio4r (2.5.8)
- * nokogiri (1.12.4)
- * oauth2 (1.4.4)
- * pagy (3.10.0)
- * paper_trail (11.1.0)
- * paper_trail-sinatra (0.8.0)
- * pg (1.2.3)
- * phonedlib (0.6.47)
- * pry (0.13.1)
- * public_suffix (4.0.6)
- * puma (5.3.2)
- * pundit (2.1.0)
- * raabro (1.4.0)
- * racc (1.5.2)
- * rack (2.2.3)
- * rack-cors (1.1.1)
- * rack-protection (2.1.0)
- * rack-test (1.1.0)
- * rake (13.0.3)
- * recaptcha (5.6.0)
- * redis (4.2.5)
- * reform (2.5.0)
- * representable (3.0.4)
- * request_store (1.5.0)
- * rest-client (2.1.0)
- * ripl (0.7.1)
- * ripl-multi_line (0.3.1)
- * ripl-rack (0.2.0)
- * rotp (6.2.0)
- * rspec (3.10.0)

```
* rspec-core (3.10.1)
* rspec-expectations (3.10.1)
* rspec-mocks (3.10.1)
* rspec-support (3.10.1)
* ruby-vips (2.1.0)
* ruby2_keywords (0.0.2)
* rubyXL (3.4.16)
* rubyzip (2.3.0)
* rufus-scheduler (3.7.0)
* sentry-ruby (4.2.0)
* sentry-ruby-core (4.2.0)
* sentry-sidekiq (4.2.0)
* shotgun (0.9.2 1cbbe4a)
* sidekiq (6.1.2)
* sidekiq-cron (1.2.0)
* sidekiq-scheduler (3.0.1)
* sidekiq_alive (2.0.4)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.21)
* sinatra-contrib (2.1.0)
* ssrf_filter (1.0.7)
* stripe (5.29.0)
* thread_safe (0.3.6)
* thwait (0.2.0)
* tilt (2.0.10)
* trailblazer (2.1.0)
* trailblazer-activity (0.12.1)
* trailblazer-activity-dsl-linear (0.4.0)
* trailblazer-context (0.4.0)
* trailblazer-developer (0.0.21)
* trailblazer-macro (2.1.6)
* trailblazer-macro-contract (2.1.1)
* trailblazer-operation (0.7.4)
* tux (0.3.0)
* tzinfo (1.2.9)
* uber (0.1.0)
* unf (0.1.4)
* unf_ext (0.0.7.7)
* zeitwerk (2.4.2)
```

2. <https://gitlab.com/tbrain-ru/aid/transactions-monitoring-service>

- a. **Description:** repo of Transaction Monitoring Service.
- b. **Structure:**

```
├── app
│   ├── commands
│   │   ├── accounts
│   │   ├── accounts_clients
│   │   ├── aid_events
│   │   ├── alerts
│   │   ├── attachments
│   │   ├── cases
│   │   ├── checkup_results
│   │   ├── checkups
│   │   │   └── core_bank
│   │   ├── clients
│   │   ├── comments
│   │   ├── documents
│   │   ├── entities
│   │   ├── events
│   │   ├── instruments
│   │   ├── misc
│   │   ├── notifications
│   │   ├── providers
│   │   ├── rules
│   │   ├── rulesets
│   │   │   └── expressions
│   │   ├── schedulers
│   │   └── tags
│   ├── contracts
│   ├── controllers
│   │   └── external
│   ├── handlers
│   ├── helpers
│   │   └── filters
│   ├── models
│   │   ├── concerns
│   │   └── dictionaries
│   ├── providers
│   │   ├── aid
│   │   │   ├── checkups
│   │   │   ├── commands
│   │   │   │   └── rules
│   │   │   │       ├── amount
│   │   │   │       ├── count_checkup
│   │   │   │       └── sum_checkup
│   │   │   ├── contracts
│   │   │   │   └── rules
│   │   └── unit21
│   │       ├── commands
```



c. **Dependencies:**

```
* aasm (5.1.1)
* activemodel (6.0.3.4)
* activerecord (6.0.3.4)
* activesupport (6.0.3.4)
* addressable (2.7.0)
* amq-protocol (2.3.2)
* blueprinter (0.23.4)
```

- * builder (3.2.4)
- * bundler (2.2.15)
- * bunny (2.17.0)
- * bunny_encrypted (0.1.0 fd45d2f)
- * carrierwave (2.1.0)
- * cmath (1.0.0)
- * coderay (1.1.3)
- * concurrent-ruby (1.1.7)
- * connection_pool (2.2.3)
- * declarative (0.0.20)
- * declarative-builder (0.1.0)
- * declarative-option (0.1.0)
- * disposable (0.4.7)
- * domain_name (0.5.20190701)
- * dotenv (2.7.6)
- * dry-auto_inject (0.7.0)
- * dry-configurable (0.11.6)
- * dry-container (0.7.2)
- * dry-core (0.4.9)
- * dry-equalizer (0.3.0)
- * dry-events (0.2.0)
- * dry-inflector (0.2.0)
- * dry-initializer (3.0.3)
- * dry-logic (1.0.7)
- * dry-monitor (0.3.2)
- * dry-schema (1.5.3)
- * dry-struct (1.3.0)
- * dry-system (0.18.1)
- * dry-types (1.4.0)
- * dry-validation (1.5.4)
- * excon (0.78.1)
- * faraday (1.3.0)
- * faraday-net_http (1.0.1)
- * ffi (1.14.2)
- * fog-aws (3.8.0)
- * fog-core (2.2.3)
- * fog-json (1.2.0)
- * fog-xml (0.1.3)
- * formatador (0.2.5)
- * hashie (4.1.0)
- * hirb (0.7.3)
- * http-accept (1.7.0)
- * http-cookie (1.0.3)
- * i18n (1.8.5)
- * ice_nine (0.11.2)
- * image_processing (1.12.1)

- * `ipaddress` (0.8.3)
- * `jwt` (2.2.2)
- * `keisan` (0.8.6)
- * `method_source` (1.0.0)
- * `mime-types` (3.3.1)
- * `mime-types-data` (3.2020.0512)
- * `mimemagic` (0.3.10)
- * `mini_magick` (4.11.0)
- * `mini_mime` (1.0.2)
- * `mini_portile2` (2.5.0)
- * `minitest` (5.14.2)
- * `multi_json` (1.15.0)
- * `multipart-post` (2.1.1)
- * `mustermann` (1.1.1)
- * `netrc` (0.11.0)
- * `nio4r` (2.5.4)
- * `nokogiri` (1.11.1)
- * `pagy` (3.7.3)
- * `paper_trail` (11.1.0)
- * `paper_trail-sinatra` (0.8.0)
- * `pg` (1.2.3)
- * `pry` (0.13.1)
- * `public_suffix` (4.0.6)
- * `puma` (5.0.2)
- * `racc` (1.5.2)
- * `rack` (2.2.3)
- * `rack-contrib` (2.2.0)
- * `rack-cors` (1.1.1)
- * `rack-protection` (2.0.8.1)
- * `rake` (12.3.3)
- * `redis` (4.2.2)
- * `reform` (2.3.3)
- * `representable` (3.0.4)
- * `request_store` (1.5.0)
- * `rest-client` (2.1.0)
- * `rmagick` (4.2.4)
- * `ruby-vips` (2.0.17)
- * `ruby2_keywords` (0.0.2)
- * `sentry-ruby` (4.2.2)
- * `sentry-ruby-core` (4.2.2)
- * `sentry-sidekiq` (4.2.1)
- * `shotgun` (0.9.2)
- * `sidekiq` (6.1.2)
- * `sidekiq_alive` (2.0.4)
- * `sinatra` (2.0.8.1)
- * `sinatra-activerecord` (2.0.21)

```
* thread_safe (0.3.6)
* tilt (2.0.10)
* trailblazer (2.1.0)
* trailblazer-activity (0.11.3)
* trailblazer-activity-dsl-linear (0.3.2)
* trailblazer-context (0.3.1)
* trailblazer-developer (0.0.16)
* trailblazer-macro (2.1.3)
* trailblazer-macro-contract (2.1.0)
* trailblazer-operation (0.6.5)
* tzinfo (1.2.7)
* uber (0.1.0)
* unf (0.1.4)
* unf_ext (0.0.7.7)
* zeitwerk (2.4.0)
```

3. <https://gitlab.com/tbrain-ru/aid/stat-service>

a. **Description:** repo of Stats Service.

b. **Structure:**

```
├── app
│   ├── controllers
│   ├── helpers
│   ├── models
│   │   ├── aid_back
│   │   ├── audit_back
│   │   ├── connections
│   │   └── stats_back
│   ├── operations
│   │   ├── clients
│   │   │   ├── cases
│   │   │   │   └── sla
│   │   │   │       └── dynamic
│   │   └── stats
│   ├── queries
│   ├── serializers
│   ├── services
│   ├── validations
│   │   └── cases
│   └── workers
├── bin
├── config
└── development
```

```
|   ├── production
|   └── test
|   └── db
|       └── migrate
|   └── lib
|       ├── auth
|       │   └── strategies
|       ├── faker
|       ├── sidekiq
|       │   └── middleware
|       └── tasks
└── system
    └── boot
```

c. Dependencies:

```
* aasm (5.1.1)
* activemodel (6.1.0.rc1)
* activerecord (6.1.0.rc1)
* activesupport (6.1.0.rc1)
* blueprinter (0.25.2)
* bond (0.5.1)
* bundler (2.1.4)
* byebug (11.1.3)
* coderay (1.1.3)
* concurrent-ruby (1.1.7)
* connection_pool (2.2.5)
* database_cleaner (2.0.0.beta f7b9246)
* database_cleaner-active_record (2.0.0.beta2)
* database_cleaner-core (2.0.0.beta2 f7b9246)
* declarative (0.0.20)
* declarative-builder (0.1.0)
* declarative-option (0.1.0)
* diff-lcs (1.4.4)
* disposable (0.4.7)
* dotenv (2.7.6)
* dry-auto_inject (0.7.0)
* dry-configurable (0.11.6)
* dry-container (0.7.2)
* dry-core (0.4.10)
* dry-equalizer (0.3.0)
* dry-events (0.2.0)
* dry-inflector (0.2.0)
* dry-initializer (3.0.4)
* dry-logic (1.0.8)
```

- * dry-monitor (0.3.2)
- * dry-schema (1.5.6)
- * dry-struct (1.3.0)
- * dry-system (0.18.1)
- * dry-types (1.4.0)
- * dry-validation (1.5.6)
- * e2mmmap (0.1.0)
- * et-orbi (1.2.6)
- * faraday (1.3.0)
- * faraday-net_http (1.0.1)
- * fugit (1.5.2)
- * hashie (4.1.0)
- * hirb (0.7.3)
- * i18n (1.8.5)
- * ice_nine (0.11.2)
- * jwt (2.2.2)
- * jwt_sessions (2.5.2)
- * method_source (1.0.0)
- * minitest (5.14.2)
- * multi_json (1.15.0)
- * multi_xml (0.6.0)
- * multipart-post (2.1.1)
- * mustermann (1.1.1)
- * nio4r (2.5.4)
- * oauth2 (1.4.4)
- * pagy (3.12.0)
- * pg (1.2.3)
- * pry (0.13.1)
- * puma (5.0.4)
- * raabro (1.4.0)
- * rack (2.2.3)
- * rack-cors (1.1.1)
- * rack-protection (2.1.0)
- * rack-test (1.1.0)
- * rake (13.0.1)
- * redis (4.2.5)
- * redis-store (1.9.0)
- * reform (2.3.3)
- * representable (3.0.4)
- * ripl (0.7.1)
- * ripl-multi_line (0.3.1)
- * ripl-rack (0.2.0)
- * rspec (3.10.0)
- * rspec-core (3.10.0)
- * rspec-expectations (3.10.0)
- * rspec-mocks (3.10.0)

```
* rspec-support (3.10.0)
* ruby2_keywords (0.0.2)
* rufus-scheduler (3.8.0)
* sentry-ruby (4.2.2)
* sentry-ruby-core (4.2.2)
* sentry-sidekiq (4.2.1)
* shotgun (0.9.2)
* sidekiq (6.3.1)
* sidekiq-scheduler (3.0.1)
* sidekiq_alive (2.1.4)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.25)
* sinatra-contrib (2.1.0)
* thwait (0.2.0)
* tilt (2.0.10)
* trailblazer (2.1.0)
* trailblazer-activity (0.11.3)
* trailblazer-activity-dsl-linear (0.3.2)
* trailblazer-context (0.3.1)
* trailblazer-developer (0.0.16)
* trailblazer-macro (2.1.3)
* trailblazer-macro-contract (2.1.0)
* trailblazer-operation (0.6.5)
* tux (0.3.0)
* tzinfo (2.0.3)
* uber (0.1.0)
* webrick (1.7.0)
* zeitwerk (2.4.1)
```

4. <https://gitlab.com/tbrain-ru/aid/submissions-service>

a. **Description:** repo of Submissions Service.

b. **Structure:**

```
|— app
|   |— commands
|   |   |— clients
|   |   |— events
|   |   |— submissions
|   |— contracts
|   |— controllers
|   |— handlers
|   |— helpers
|   |— models
```

```
|   ├── serializers
|   ├── services
|   └── workers
|── bin
|── config
|── db
|   └── migrate
|── lib
|   └── tasks
└── system
    └── boot
```

c. Dependencies:

```
* aa_cipher (0.0.25 b4984cb)
* activemodel (6.0.3.5)
* activerecord (6.0.3.5)
* activesupport (6.0.3.5)
* amq-protocol (2.3.2)
* argon2-kdf (0.1.6)
* blind_index (2.2.0)
* blueprinter (0.25.1)
* bundler (2.1.4)
* bunny (2.17.0)
* bunny_encrypted (0.1.0 fd45d2f)
* byebug (11.1.3)
* coderay (1.1.3)
* concurrent-ruby (1.1.8)
* connection_pool (2.2.3)
* declarative (0.0.20)
* declarative-builder (0.1.0)
* declarative-option (0.1.0)
* disposable (0.4.7)
* dotenv (2.7.6)
* dry-configurable (0.11.6)
* dry-container (0.7.2)
* dry-core (0.4.9)
* dry-equalizer (0.3.0)
* dry-inflector (0.2.0)
* dry-initializer (3.0.4)
* dry-logic (1.0.8)
* dry-monads (1.3.5)
* dry-schema (1.5.6)
* dry-types (1.4.0)
* dry-validation (1.5.6)
```

- * `elasticsearch` (7.9.0)
- * `elasticsearch-api` (7.9.0)
- * `elasticsearch-model` (7.1.1)
- * `elasticsearch-transport` (7.9.0)
- * `faraday` (1.1.0)
- * `hashie` (4.1.0)
- * `hirb` (0.7.3)
- * `i18n` (1.8.9)
- * `method_source` (1.0.0)
- * `minitest` (5.14.4)
- * `multi_json` (1.15.0)
- * `multipart-post` (2.1.1)
- * `mustermann` (1.1.1)
- * `nio4r` (2.5.8)
- * `pg` (1.2.3)
- * `pry` (0.13.1)
- * `puma` (5.3.2)
- * `rack` (2.2.3)
- * `rack-contrib` (2.2.0)
- * `rack-cors` (1.1.1)
- * `rack-protection` (2.1.0)
- * `rake` (13.0.1)
- * `redis` (4.2.5)
- * `reform` (2.3.3)
- * `representable` (3.0.4)
- * `ruby2_keywords` (0.0.2)
- * `sentry-ruby` (4.2.2)
- * `sentry-ruby-core` (4.2.2)
- * `sentry-sidekiq` (4.2.1)
- * `sidekiq` (6.1.3)
- * `sidekiq_alive` (2.0.4)
- * `sinatra` (2.1.0)
- * `sinatra-activerecord` (2.0.21)
- * `sinatra-contrib` (2.1.0)
- * `thread_safe` (0.3.6)
- * `tilt` (2.0.10)
- * `trailblazer` (2.1.0)
- * `trailblazer-activity` (0.11.3)
- * `trailblazer-activity-dsl-linear` (0.3.2)
- * `trailblazer-context` (0.3.1)
- * `trailblazer-developer` (0.0.16)
- * `trailblazer-macro` (2.1.3)
- * `trailblazer-macro-contract` (2.1.0)
- * `trailblazer-operation` (0.6.5)
- * `tzinfo` (1.2.9)
- * `uber` (0.1.0)

* zeitwerk (2.4.2)

5. <https://gitlab.com/tbrain-ru/aid/checkups-service>

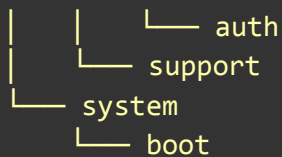
a. **Description:** repo of Checkups Service.

b. **Structure:**

```
├── app
│   ├── commands
│   │   ├── admin
│   │   │   ├── checkup_kinds
│   │   │   ├── checkup_kinds_clients
│   │   │   └── clients_providers
│   │   ├── aid
│   │   │   ├── accounts
│   │   │   ├── accounts_clients
│   │   │   └── clients
│   │   ├── audit_configs
│   │   ├── audit_configs_checkup_kinds
│   │   ├── audits
│   │   ├── audits_checkups
│   │   ├── checkup_forms
│   │   ├── checkups
│   │   ├── common
│   │   │   ├── checkup_kinds_clients
│   │   │   └── clients_providers
│   │   ├── events
│   │   ├── misc
│   │   ├── ongoing_configs
│   │   └── provider_features
│   │       └── comply_advantage
│   ├── contracts
│   │   ├── accounts
│   │   ├── accounts_clients
│   │   ├── audit_configs
│   │   ├── audit_configs_checkup_kinds
│   │   ├── audits
│   │   ├── checkup_kinds
│   │   │   ├── aid
│   │   │   ├── comply_advantage
│   │   │   ├── elliptic
│   │   │   ├── maxmind
│   │   │   └── unit21
│   └── checkups
```



```
|
|
|   ├── clients
|   ├── events
|   ├── ongoing_configs
|   ├── provider_features
|   |   └── comply_advantage
|   ├── controllers
|   |   ├── admin
|   |   └── client_management
|   ├── executors
|   |   ├── aid
|   |   ├── comply_advantage
|   |   ├── elliptic
|   |   ├── maxmind
|   |   └── unit21
|   ├── handlers
|   ├── helpers
|   ├── models
|   ├── policies
|   ├── serializers
|   ├── services
|   |   ├── aid_data
|   |   ├── checkups
|   |   ├── documents
|   |   ├── http
|   |   ├── providers
|   |   |   ├── comply_advantage
|   |   |   ├── elliptic
|   |   |   ├── maxmind
|   |   |   └── unit21
|   |   ├── rabbit
|   |   ├── stubs
|   |   └── submissions
|   └── workers
|
| ├── bin
| ├── config
| ├── db
| |   └── migrate
| ├── lib
| |   ├── auth
| |   |   └── strategies
| |   ├── sidekiq
| |   |   └── middleware
| |   └── tasks
| ├── public
| ├── spec
| └── shared
```



c. Dependencies:

```
* a_command (0.1.4 a736c04)
* aasm (5.2.0)
* activemodel (6.1.4.4)
* activerecord (6.1.4.4)
* activesupport (6.1.4.4)
* amq-protocol (2.3.2)
* blueprinter (0.25.3)
* bond (0.5.1)
* bundler (2.3.5)
* bunny (2.19.0)
* bunny_encrypted (0.1.0 fd45d2f)
* byebug (11.1.3)
* coderay (1.1.3)
* concurrent-ruby (1.1.9)
* connection_pool (2.2.5)
* database_cleaner (2.0.1 068656a)
* database_cleaner-active_record (2.0.0)
* database_cleaner-core (2.0.1 068656a)
* diff-lcs (1.4.4)
* domain_name (0.5.20190701)
* dotenv (2.7.6)
* dry-auto_inject (0.7.0)
* dry-configurable (0.12.1)
* dry-container (0.7.2)
* dry-core (0.5.0)
* dry-equalizer (0.3.0)
* dry-events (0.2.0)
* dry-inflector (0.2.0)
* dry-initializer (3.0.4)
* dry-logic (1.1.0)
* dry-monitor (0.3.2)
* dry-schema (1.6.1)
* dry-struct (1.4.0)
* dry-system (0.18.1)
* dry-types (1.5.1)
* dry-validation (1.6.0)
* e2mmap (0.1.0)
* et-orbi (1.2.6)
```

- * factory_bot (6.2.0)
- * faker (2.20.0)
- * faraday (1.3.0)
- * faraday-net_http (1.0.1)
- * fugit (1.5.2)
- * http-accept (1.7.0)
- * http-cookie (1.0.4)
- * i18n (1.8.11)
- * ice_nine (0.11.2)
- * jwt (2.2.2)
- * jwt_sessions (2.5.2)
- * method_source (1.0.0)
- * mime-types (3.4.1)
- * mime-types-data (3.2022.0105)
- * minitest (5.15.0)
- * multi_json (1.15.0)
- * multi_xml (0.6.0)
- * multipart-post (2.1.1)
- * mustermann (1.1.1)
- * netrc (0.11.0)
- * nio4r (2.5.8)
- * oauth2 (1.4.7)
- * pagy (5.8.1)
- * pg (1.2.3)
- * pry (0.14.0)
- * puma (5.5.2)
- * pundit (2.1.0)
- * raabro (1.4.0)
- * rack (2.2.3)
- * rack-cors (1.1.1)
- * rack-protection (2.1.0)
- * rack-test (1.1.0)
- * rake (13.0.6)
- * rbtree (0.4.4)
- * redis (4.6.0)
- * rest-client (2.1.0)
- * ripl (0.7.1)
- * ripl-multi_line (0.3.1)
- * ripl-rack (0.2.0)
- * rspec (3.10.0)
- * rspec-core (3.10.1)
- * rspec-expectations (3.10.1)
- * rspec-mocks (3.10.2)
- * rspec-support (3.10.2)
- * ruby2_keywords (0.0.5)
- * rufus-scheduler (3.8.1)

```
* sentry-ruby (4.3.1)
* sentry-ruby-core (4.3.1)
* set (1.0.2)
* shotgun (0.9.2)
* sidekiq (6.4.1)
* sidekiq-scheduler (3.0.1)
* sidekiq_alive (2.1.4)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.22)
* sinatra-contrib (2.1.0)
* sorted_set (1.0.3)
* thwait (0.2.0)
* tilt (2.0.10)
* tux (0.3.0)
* tzinfo (2.0.4)
* unf (0.1.4)
* unf_ext (0.0.8)
* webrick (1.7.0)
* zeitwerk (2.5.3)
```

6. <https://gitlab.com/tbrain-ru/aid/aid-scoring-service>

a. **Description:** repo for Scoring Service.

b. **Structure:**

```
|— app
|   |— commands
|   |   |— aid
|   |   |   |— accounts
|   |   |   |— clients
|   |   |— case_evaluations
|   |   |   |— by_case
|   |   |— case_section_evaluations
|   |   |   |— by_case
|   |   |— case_section_subject_evaluations
|   |   |— events
|   |   |— field_submission_evaluations
|   |   |   |— by_form_submission
|   |   |— misc
|   |   |— rules
|   |   |   |— by_form_configuration
|   |   |— scoring_models
|   |   |   |— by_form_configuration
|   |   |— submission_evaluations
```

```

├── by_form_submission
├── contracts
├── controllers
├── handlers
├── helpers
├── models
├── serializers
├── services
├── ┬── aid_scoring
├── │ ├── core
├── │ ├── rabbit
├── │ ├── stubs
├── │ │ └── files
├── │ └── submissions
├── workers
├── bin
├── config
├── db
├── └── migrate
├── lib
├── │ ├── auth
├── │ │ └── strategies
├── │ ├── elasticrules
├── │ ├── sidekiq
├── │ │ └── middleware
├── │ └── tasks
├── spec
├── │ ├── factories
├── │ ├── requests
├── │ ├── shared
├── │ │ └── auth
├── │ └── support
├── system
├── └── boot

```

c. Dependencies:

```

* a_command (0.1.2 f32b136)
* aasm (5.2.0)
* activemodel (6.0.3.4)
* activerecord (6.0.3.4)
* activesupport (6.0.3.4)
* amq-protocol (2.3.2)
* blueprinter (0.25.1)
* bunny (2.19.0)

```

- * bunny_encrypted (0.1.0 fd45d2f)
- * byebug (11.1.3)
- * coderay (1.1.3)
- * concurrent-ruby (1.1.7)
- * connection_pool (2.2.5)
- * database_cleaner-active_record (2.0.1)
- * database_cleaner-core (2.0.1)
- * diff-lcs (1.4.4)
- * dotenv (2.7.6)
- * dry-auto_inject (0.7.0)
- * dry-configurable (0.11.6)
- * dry-container (0.7.2)
- * dry-core (0.4.9)
- * dry-equalizer (0.3.0)
- * dry-events (0.2.0)
- * dry-inflector (0.2.0)
- * dry-initializer (3.0.4)
- * dry-logic (1.0.8)
- * dry-monitor (0.3.2)
- * dry-schema (1.5.6)
- * dry-struct (1.3.0)
- * dry-system (0.18.1)
- * dry-types (1.4.0)
- * dry-validation (1.6.0)
- * e2mmap (0.1.0)
- * et-orbi (1.2.6)
- * factory_bot (6.2.0)
- * faker (2.19.0)
- * faraday (1.3.0)
- * faraday-net_http (1.0.1)
- * fugit (1.5.2)
- * i18n (1.8.5)
- * ice_nine (0.11.2)
- * jwt (2.3.0)
- * method_source (1.0.0)
- * minitest (5.14.2)
- * multi_json (1.15.0)
- * multipart-post (2.1.1)
- * mustermann (1.1.1)
- * nio4r (2.5.8)
- * pagy (3.14.0)
- * pg (1.2.3)
- * pry (0.13.1)
- * puma (5.3.2)
- * raabro (1.4.0)
- * rack (2.2.3)

```
* rack-cors (1.1.1)
* rack-protection (2.1.0)
* rack-test (1.1.0)
* rake (13.0.1)
* rbtree (0.4.4)
* redis (4.5.1)
* rspec (3.10.0)
* rspec-core (3.10.1)
* rspec-expectations (3.10.1)
* rspec-mocks (3.10.2)
* rspec-support (3.10.3)
* ruby2_keywords (0.0.2)
* rufus-scheduler (3.8.0)
* sentry-ruby (4.8.1)
* sentry-ruby-core (4.8.1)
* sentry-sidekiq (4.8.1)
* set (1.0.2)
* shotgun (0.9.2)
* sidekiq (6.3.1)
* sidekiq-scheduler (3.0.1)
* sidekiq_alive (2.1.4)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.23)
* sinatra-contrib (2.1.0)
* sorted_set (1.0.3)
* thread_safe (0.3.6)
* thwait (0.2.0)
* tilt (2.0.10)
* tzinfo (1.2.8)
* webrick (1.7.0)
* zeitwerk (2.4.1)
```

7. <https://gitlab.com/tbrain-ru/aid/audit-service>

a. **Description:** repo for Audit Service.

b. **Structure:**

```
├── app
│   ├── commands
│   │   ├── events
│   │   ├── misc
│   │   └── records
│   ├── contracts
│   └── controllers
```

```
|   |— handlers
|   |— helpers
|   |— models
|   |— queries
|   |— serializers
|   |— services
|   |   |— audit
|   |   |— rabbit
|— bin
|— config
|— db
|   |— migrate
|— lib
|   |— tasks
|— spec
|— system
|   |— boot
```

c. Dependencies:

```
* a_command (0.1.0 df22da3)
* activemodel (6.0.4.1)
* activerecord (6.0.4.1)
* activesupport (6.0.4.1)
* amq-protocol (2.3.2)
* blueprinter (0.25.1)
* bunny (2.19.0)
* bunny_encrypted (0.1.0 fd45d2f)
* byebug (11.1.3)
* coderay (1.1.3)
* concurrent-ruby (1.1.9)
* diff-lcs (1.4.4)
* dotenv (2.7.6)
* dry-auto_inject (0.7.0)
* dry-configurable (0.11.6)
* dry-container (0.7.2)
* dry-core (0.4.9)
* dry-equalizer (0.3.0)
* dry-events (0.2.0)
* dry-inflector (0.2.0)
* dry-initializer (3.0.4)
* dry-logic (1.0.8)
* dry-monitor (0.3.2)
* dry-schema (1.5.6)
* dry-struct (1.3.0)
```



```
* dry-system (0.18.1)
* dry-types (1.4.0)
* dry-validation (1.6.0)
* faraday (1.1.0)
* i18n (1.8.10)
* ice_nine (0.11.2)
* method_source (1.0.0)
* minitest (5.14.4)
* multi_json (1.15.0)
* multipart-post (2.1.1)
* mustermann (1.1.1)
* nio4r (2.5.8)
* pagy (3.12.0)
* pg (1.2.3)
* pry (0.13.1)
* puma (5.3.2)
* rack (2.2.3)
* rack-cors (1.1.1)
* rack-protection (2.1.0)
* rake (13.0.1)
* rbtree (0.4.4)
* rspec (3.10.0)
* rspec-core (3.10.1)
* rspec-expectations (3.10.1)
* rspec-mocks (3.10.2)
* rspec-support (3.10.2)
* ruby2_keywords (0.0.2)
* sentry-ruby (4.2.2)
* sentry-ruby-core (4.2.2)
* set (1.0.1)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.22)
* sinatra-contrib (2.1.0)
* sorted_set (1.0.3)
* thread_safe (0.3.6)
* tilt (2.0.10)
* tzinfo (1.2.9)
* zeitwerk (2.4.2)
```

8. https://gitlab.com/tbrain-ru/aid/documents_service

a. **Description:** repo for Documents Service.

b. **Structure:**

```
|— app
|   |— commands
|   |   |— aid
|   |   |   |— accounts
|   |   |   |— accounts_clients
|   |   |   |— clients
|   |   |   |— customers
|   |   |— callbacks
|   |   |— documents
|   |   |— events
|   |   |— public_documents
|   |— contracts
|   |   |— accounts
|   |   |— accounts_clients
|   |   |— clients
|   |   |— customers
|   |   |— documents
|   |   |— events
|   |   |— public_documents
|   |— controllers
|   |— handlers
|   |— helpers
|   |— models
|   |— policies
|   |— serializers
|   |— services
|   |   |— aid_data
|   |   |— rabbit
|   |   |— stubs
|   |— uploaders
|— bin
|— config
|— db
|   |— migrate
|— lib
|   |— auth
|   |   |— strategies
|   |— safe_upload
|   |— tasks
|— system
|   |— boot
```

c. **Dependencies:**

```
* aasm (5.2.0)
```

- * activemodel (6.1.3.1)
- * activerecord (6.1.3.1)
- * activesupport (6.1.3.1)
- * addressable (2.8.0)
- * amq-protocol (2.3.2)
- * aws-eventstream (1.1.1)
- * aws-partitions (1.447.0)
- * aws-sdk-core (3.114.0)
- * aws-sdk-kms (1.43.0)
- * aws-sdk-s3 (1.93.1)
- * aws-sigv4 (1.2.3)
- * blueprinter (0.25.3)
- * bond (0.5.1)
- * bunny (2.17.0)
- * bunny_encrypted (0.1.0 fd45d2f)
- * bybug (11.1.3)
- * carrierwave (2.2.1)
- * carrierwave-aws (1.5.0)
- * coderay (1.1.3)
- * concurrent-ruby (1.1.8)
- * database_cleaner (2.0.1 068656a)
- * database_cleaner-active_record (2.0.0)
- * database_cleaner-core (2.0.1 068656a)
- * declarative (0.0.20)
- * declarative-builder (0.1.0)
- * declarative-option (0.1.0)
- * diff-lcs (1.4.4)
- * disposable (0.4.7)
- * dotenv (2.7.6)
- * dry-auto_inject (0.7.0)
- * dry-configurable (0.12.1)
- * dry-container (0.7.2)
- * dry-core (0.5.0)
- * dry-equalizer (0.3.0)
- * dry-events (0.2.0)
- * dry-inflector (0.2.0)
- * dry-initializer (3.0.4)
- * dry-logic (1.1.0)
- * dry-monitor (0.3.2)
- * dry-schema (1.6.1)
- * dry-struct (1.4.0)
- * dry-system (0.18.1)
- * dry-types (1.5.1)
- * dry-validation (1.6.0)
- * faraday (1.3.0)
- * faraday-net_http (1.0.1)

- * ffi (1.15.0)
- * hashie (4.1.0)
- * hirb (0.7.3)
- * i18n (1.8.10)
- * ice_nine (0.11.2)
- * image_processing (1.12.1)
- * jmespath (1.4.0)
- * jwt (2.2.2)
- * jwt_sessions (2.5.2)
- * marcel (1.0.1)
- * method_source (1.0.0)
- * mimemagic (0.3.10)
- * mini_magick (4.11.0)
- * mini_mime (1.1.0)
- * mini_portile2 (2.6.1)
- * minitest (5.14.4)
- * multi_json (1.15.0)
- * multi_xml (0.6.0)
- * multipart-post (2.1.1)
- * mustermann (1.1.1)
- * nio4r (2.5.8)
- * nokogiri (1.12.4)
- * oauth2 (1.4.7)
- * pagy (3.12.0)
- * pg (1.2.3)
- * pry (0.14.0)
- * public_suffix (4.0.6)
- * puma (5.4.0)
- * pundit (2.1.0)
- * racc (1.5.2)
- * rack (2.2.3)
- * rack-cors (1.1.1)
- * rack-protection (2.1.0)
- * rack-test (1.1.0)
- * rake (13.0.3)
- * reform (2.5.0)
- * representable (3.0.4)
- * ripl (0.7.1)
- * ripl-multi_line (0.3.1)
- * ripl-rack (0.2.0)
- * rspec (3.10.0)
- * rspec-core (3.10.1)
- * rspec-expectations (3.10.1)
- * rspec-mocks (3.10.2)
- * rspec-support (3.10.2)
- * ruby-vips (2.1.0)

```
* ruby2_keywords (0.0.4)
* sentry-ruby (4.3.1)
* sentry-ruby-core (4.3.1)
* shotgun (0.9.2)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.22)
* sinatra-contrib (2.1.0)
* ssrf_filter (1.0.7)
* tilt (2.0.10)
* trailblazer (2.1.0)
* trailblazer-activity (0.12.1)
* trailblazer-activity-dsl-linear (0.4.0)
* trailblazer-context (0.4.0)
* trailblazer-developer (0.0.21)
* trailblazer-macro (2.1.6)
* trailblazer-macro-contract (2.1.1)
* trailblazer-operation (0.7.4)
* tux (0.3.0)
* tzinfo (2.0.4)
* uber (0.1.0)
* zeitwerk (2.4.2)
```

9. <https://gitlab.com/tbrain-ru/aid/user-behaviour-tracking-service>

a. **Description:** repo of User Behavior Tracking service.

b. **Structure:**

```
├── app
│   ├── contracts
│   ├── controllers
│   ├── handlers
│   ├── helpers
│   │   └── events
│   ├── models
│   ├── operations
│   │   ├── clients
│   │   ├── events
│   │   ├── stats
│   │   └── submissions
│   ├── serializers
│   ├── services
│   └── workers
├── bin
└── config
```

```
|— db
|   |— migrate
|— lib
|   |— auth
|— spec
|— system
|   |— boot
```

c. Dependencies:

```
* activemodel (6.0.3.2)
* activerecord (6.0.3.2)
* activesupport (6.0.3.2)
* amq-protocol (2.3.2)
* bcrypt (3.1.15)
* blueprinter (0.23.4)
* bond (0.5.1)
* bunny (2.17.0)
* bunny_encrypted (0.1.0 fd45d2f)
* byebug (11.1.3)
* coderay (1.1.3)
* concurrent-ruby (1.1.7)
* connection_pool (2.2.3)
* declarative (0.0.20)
* declarative-builder (0.1.0)
* declarative-option (0.1.0)
* disposable (0.4.7)
* dotenv (2.7.6)
* dry-auto_inject (0.7.0)
* dry-configurable (0.12.1)
* dry-container (0.7.2)
* dry-core (0.5.0)
* dry-equalizer (0.3.0)
* dry-events (0.2.0)
* dry-inflector (0.2.0)
* dry-initializer (3.0.4)
* dry-logic (1.1.0)
* dry-monitor (0.3.2)
* dry-schema (1.6.1)
* dry-struct (1.4.0)
* dry-system (0.18.1)
* dry-types (1.5.1)
* dry-validation (1.6.0)
* faraday (1.3.0)
```

```
* faraday-net_http (1.0.1)
* hashie (4.1.0)
* hirb (0.7.3)
* i18n (1.8.5)
* ice_nine (0.11.2)
* json (2.3.1)
* jwt (2.2.2)
* method_source (1.0.0)
* minitest (5.14.1)
* multipart-post (2.1.1)
* mustermann (1.1.1)
* nio4r (2.5.7)
* pg (0.20.0)
* pry (0.13.1)
* puma (5.2.2)
* rack (2.2.3)
* rack-cors (1.1.1)
* rack-protection (2.0.8.1)
* rack-test (1.1.0)
* rake (13.0.1)
* redis (4.2.1)
* reform (2.3.3)
* representable (3.0.4)
* require_all (3.0.0)
* ripl (0.7.1)
* ripl-multi_line (0.3.1)
* ripl-rack (0.2.0)
* ruby2_keywords (0.0.2)
* sentry-ruby (4.2.2)
* sentry-ruby-core (4.2.2)
* sentry-sidekiq (4.2.1)
* shotgun (0.9.2)
* sidekiq (6.1.1)
* sidekiq_alive (2.0.3)
* sinatra (2.0.8.1)
* sinatra-activerecord (2.0.18)
* sinatra-cross_origin (0.4.0)
* thread_safe (0.3.6)
* tilt (2.0.10)
* trailblazer (2.1.0)
* trailblazer-activity (0.11.3)
* trailblazer-activity-dsl-linear (0.3.2)
* trailblazer-context (0.3.1)
* trailblazer-developer (0.0.16)
* trailblazer-macro (2.1.3)
* trailblazer-macro-contract (2.1.0)
```

```
* trailblazer-operation (0.6.5)
* tux (0.3.0)
* tzinfo (1.2.7)
* uber (0.1.0)
* zeitwerk (2.4.0)
```

10. <https://gitlab.com/tbrain-ru/aid/passport-service>

a. **Description:** repo for GraphQL API of Passport Service.

b. **Structure:**

```
|— app
|   |— commands
|   |   |— aid
|   |   |   |— accounts
|   |   |— events
|   |   |— misc
|   |— contracts
|   |— controllers
|   |— handlers
|   |— helpers
|   |— models
|   |— queries
|   |— serializers
|   |— services
|   |   |— passport
|   |   |— passport_data
|   |   |— rabbit
|— bin
|— config
|— db
|   |— migrate
|— lib
|   |— auth
|   |   |— strategies
|   |— tasks
|— spec
|— system
|   |— boot
```

c. **Dependencies:**

```
* a_command (0.1.0 df22da3)
```


- * activemodel (6.1.2.1)
- * activerecord (6.1.2.1)
- * activesupport (6.1.2.1)
- * amq-protocol (2.3.2)
- * blueprinter (0.25.1)
- * bunny (2.19.0)
- * bunny_encrypted (0.1.0 fd45d2f)
- * byebug (11.1.3)
- * coderay (1.1.3)
- * concurrent-ruby (1.1.9)
- * diff-lcs (1.4.4)
- * dotenv (2.7.6)
- * dry-auto_inject (0.7.0)
- * dry-configurable (0.11.6)
- * dry-container (0.7.2)
- * dry-core (0.4.9)
- * dry-equalizer (0.3.0)
- * dry-events (0.2.0)
- * dry-inflector (0.2.0)
- * dry-initializer (3.0.4)
- * dry-logic (1.0.8)
- * dry-monitor (0.3.2)
- * dry-schema (1.5.6)
- * dry-struct (1.3.0)
- * dry-system (0.18.1)
- * dry-types (1.4.0)
- * dry-validation (1.6.0)
- * faraday (1.1.0)
- * graphql (1.12.16)
- * i18n (1.8.10)
- * ice_nine (0.11.2)
- * jwt (2.2.3)
- * method_source (1.0.0)
- * minitest (5.14.4)
- * multi_json (1.15.0)
- * multipart-post (2.1.1)
- * mustermann (1.1.1)
- * nio4r (2.5.8)
- * pagy (3.12.0)
- * pg (1.2.3)
- * pry (0.13.1)
- * puma (5.3.2)
- * rack (2.2.3)
- * rack-contrib (2.3.0)
- * rack-cors (1.1.1)
- * rack-protection (2.1.0)

```
* rake (13.0.1)
* rbtree (0.4.4)
* rspec (3.10.0)
* rspec-core (3.10.1)
* rspec-expectations (3.10.1)
* rspec-mocks (3.10.2)
* rspec-support (3.10.2)
* ruby2_keywords (0.0.2)
* sentry-ruby (4.2.2)
* sentry-ruby-core (4.2.2)
* set (1.0.1)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.22)
* sinatra-contrib (2.1.0)
* sorted_set (1.0.3)
* tilt (2.0.10)
* tzinfo (2.0.4)
* zeitwerk (2.4.2)
```

11. <https://gitlab.com/tbrain-ru/aid/passport-data-service>

a. **Description:** repo for geo-distributed part of Passport Service.

b. **Structure:**

```
├── app
│   ├── commands
│   │   ├── accounts_business_passports
│   │   ├── aid
│   │   │   ├── accounts
│   │   │   └── onboardings
│   │   ├── attachments
│   │   ├── business_partners
│   │   ├── business_passport_addresses
│   │   ├── business_passport_invites
│   │   ├── business_passports
│   │   ├── company_ownerships
│   │   ├── contacts
│   │   ├── cryptocurrency_wallets
│   │   ├── departments
│   │   ├── document_sections
│   │   ├── employments
│   │   ├── events
│   │   ├── family_members
│   │   └── financial_accounts
```

```
|
|
|   ├── financial_statements
|   ├── graphql
|   ├── idenfify_data
|   ├── idenfify_verifications
|   ├── identity_documents
|   ├── management_team_members
|   ├── misc
|   ├── pep_family_members
|   ├── personal_passport_addresses
|   ├── personal_passports
|   ├── shareholders
|   ├── taxations
|   ├── turnover_values
|   └── ubo_shareholders
|
|── contracts
|── controllers
|── graphql
|   ├── mutations
|   │   └── idenfify_verifications
|   ├── queries
|   └── types
|── handlers
|── helpers
|── models
|── queries
|── serializers
|── services
|   ├── billing
|   ├── core
|   ├── documents
|   ├── idenfify
|   ├── notifications
|   ├── passport_data
|   ├── rabbit
|   └── submissions
|── workers
|
|── bin
|── config
|── db
|   └── migrate
|── lib
|   ├── sidekiq
|   │   └── middleware
|   └── tasks
|── spec
|── system
```

└─ boot

c. Dependencies:

```
* a_command (0.1.1 3f5a333)
* aasm (5.2.0)
* activemodel (6.0.3.4)
* activerecord (6.0.3.4)
* activesupport (6.0.3.4)
* amq-protocol (2.3.2)
* blueprinter (0.25.1)
* bunny (2.19.0)
* bunny_encrypted (0.1.0 fd45d2f)
* byebug (11.1.3)
* coderay (1.1.3)
* concurrent-ruby (1.1.7)
* connection_pool (2.2.5)
* diff-lcs (1.4.4)
* dotenv (2.7.6)
* dry-auto_inject (0.7.0)
* dry-configurable (0.11.6)
* dry-container (0.7.2)
* dry-core (0.4.9)
* dry-equalizer (0.3.0)
* dry-events (0.2.0)
* dry-inflector (0.2.0)
* dry-initializer (3.0.4)
* dry-logic (1.0.8)
* dry-monitor (0.3.2)
* dry-schema (1.5.6)
* dry-struct (1.3.0)
* dry-system (0.18.1)
* dry-types (1.4.0)
* dry-validation (1.6.0)
* e2mmap (0.1.0)
* et-orbi (1.2.6)
* faraday (1.1.0)
* fugit (1.5.2)
* graphql (1.12.16)
* i18n (1.8.5)
* ice_nine (0.11.2)
* method_source (1.0.0)
* minitest (5.14.2)
* multi_json (1.15.0)
* multipart-post (2.1.1)
```

```
* mustermann (1.1.1)
* nio4r (2.5.4)
* pagy (3.12.0)
* pg (1.2.3)
* pry (0.13.1)
* puma (5.0.4)
* raabro (1.4.0)
* rack (2.2.3)
* rack-contrib (2.3.0)
* rack-cors (1.1.1)
* rack-protection (2.1.0)
* rake (13.0.1)
* rbtree (0.4.4)
* redis (4.5.1)
* rspec (3.10.0)
* rspec-core (3.10.1)
* rspec-expectations (3.10.1)
* rspec-mocks (3.10.2)
* rspec-support (3.10.2)
* ruby2_keywords (0.0.2)
* rufus-scheduler (3.8.0)
* sentry-ruby (4.2.2)
* sentry-ruby-core (4.2.2)
* sentry-sidekiq (4.2.1)
* set (1.0.1)
* shotgun (0.9.2)
* sidekiq (6.3.1)
* sidekiq-scheduler (3.0.1)
* sidekiq_alive (2.1.4)
* sinatra (2.1.0)
* sinatra-activerecord (2.0.22)
* sinatra-contrib (2.1.0)
* sorted_set (1.0.3)
* thread_safe (0.3.6)
* thwait (0.2.0)
* tilt (2.0.10)
* tzinfo (1.2.8)
* webrick (1.7.0)
* zeitwerk (2.4.1)
```

Golang

12. <https://gitlab.com/tbrain-ru/aid/auth-service>
 - a. **Description:** repo of Auth Service.

b. **Structure:**

```
|— bin
|— cmd
|   |— consumer
|   |   |— aid_clients
|   |— migrator
|   |— mocks
|   |   |— clos_mock
|   |   |— rabbit_events
|   |   |   |— authenticated
|   |   |   |   |— worker
|   |   |— rpc_client_server
|   |— rpc_service
|   |— service
|— db
|   |— migrate
|— internal
|   |— app
|   |   |— server
|   |— clients
|   |   |— accounts
|   |   |   |— entity
|   |   |   |— usecase
|   |— connectors
|   |   |— postgres
|   |   |— rabbit
|   |   |— redis
|   |— http
|   |   |— entity
|   |   |— handlers
|   |   |   |— oauth
|   |   |   |   |— authorize
|   |   |   |   |— introspect
|   |   |   |   |— jwt_session_list
|   |   |   |   |— jwt_session_logout
|   |   |   |   |— jwt_session_logout_all
|   |   |   |   |— token
|   |   |   |   |— user_info
|   |   |   |— web
|   |   |       |— ap_callback
|   |   |       |— api_login
|   |   |       |— api_redirect
|   |   |       |— error
|   |   |       |— gl_callback
|   |   |       |— health_check
```



```
github.com/jackc/pgx/v4 v4.1.2
github.com/jmoiron/sqlx v1.3.0
github.com/lib/pq v1.3.0
github.com/mitchellh/mapstructure v1.1.2
github.com/mssola/user_agent v0.5.3
github.com/nyaruka/phonenumbers v1.0.72
github.com/onrik/logrus v0.8.0
github.com/pkg/errors v0.9.1
github.com/pquerna/otp v1.2.0
github.com/sirupsen/logrus v1.8.1
github.com/spf13/cobra v1.0.0
github.com/spf13/viper v1.4.0
github.com/streadway/amqp v1.0.0
github.com/stretchr/testify v1.7.0
github.com/ulule/limiter/v3 v3.8.0
gitlab.com/tbrain-ru/gems/go-rabbit
v0.0.10-0.20220401154256-73657e05e74e
golang.org/x/oauth2 v0.0.0-20200107190931-bf48bf16ab8d
```

13. <https://gitlab.com/tbrain-ru/services/tfa-service>

- a. **Description:** repo of 2FA Service.
- b. **Structure:** plain (no folders)
- c. **Dependencies:**

```
github.com/fsnotify/fsnotify v1.4.9
github.com/getsentry/sentry-go v0.10.0
github.com/go-redis/redis v6.15.8+incompatible
github.com/mitchellh/mapstructure v1.3.0
github.com/pelletier/go-toml v1.8.0
github.com/spf13/afero v1.2.2
github.com/spf13/cast v1.3.1
github.com/spf13/jwalterweatherman v1.1.0
github.com/spf13/pflag v1.0.5
github.com/spf13/viper v1.7.0
github.com/streadway/amqp v1.0.0
gitlab.com/tbrain-ru/gems/go-rabbit v0.0.5
golang.org/x/sys v0.0.0-20200519105757-fe76b779f299
gopkg.in/ini.v1 v1.56.0
```

14. <https://gitlab.com/tbrain-ru/services/notification-service>

a. **Description:** repo of Notifications Service.

b. **Structure:**

```
|— cmd
|   |— phantom
|   |— worker
|— internal
|   |— clients
|   |   |— email
|   |   |— twilio
|   |— connectors
|   |   |— rabbit
|   |— pkg
|   |   |— entity
|   |   |— notify
|   |— rabbit
|   |   |— notify_worker
|   |— utils
|   |   |— deps
|   |   |— env
|   |   |— shutdown
|   |   |— templates
|— templates
|   |— aid
|   |   |— email
|   |   |— sms
|   |   |— voice
|   |— arivalbank
|   |   |— email
|   |   |— sms
|   |   |— voice
|   |— passport
|   |   |— email
|— test
```

c. **Dependencies:**

```
github.com/Scumfunk/mailiak v1.2.1-0.20200527223016-ddeddf0f8d99
github.com/fsnotify/fsnotify v1.4.9
github.com/getsentry/sentry-go v0.10.0
github.com/golang/protobuf v1.4.2
github.com/inconshreveable/log15
v0.0.0-20200109203555-b30bc20e4fd1
```

```
github.com/isayme/go-amqp-reconnect
v0.0.0-20210303120416-fc811b0bcda2
github.com/kevinburke/go-types
v0.0.0-20200309064045-f2d4aea18a7a
github.com/kevinburke/go.uuid v1.2.0
github.com/kevinburke/rest v0.0.0-20200429221318-0d2892b400f8
github.com/kevinburke/twilio-go
v0.0.0-20190630185733-fe05957cdaf8
github.com/mattn/go-colorable v0.1.6
github.com/mitchellh/mapstructure v1.3.1
github.com/onrik/logrus v0.9.0
github.com/pelletier/go-toml v1.8.0
github.com/sirupsen/logrus v1.2.0
github.com/spf13/afero v1.2.2
github.com/spf13/cast v1.3.1
github.com/spf13/jwalterweatherman v1.1.0
github.com/spf13/pflag v1.0.5
github.com/spf13/viper v1.7.0
github.com/streadway/amqp v1.0.0
github.com/ttacon/builder v0.0.0-20170518171403-c099f663e1c2
github.com/ttacon/libphonenumber v1.1.0
gitlab.com/tbrain-ru/gems/go-rabbit v0.0.8
golang.org/x/net v0.0.0-20200520182314-0ba52f642ac2
golang.org/x/sync v0.0.0-20200317015054-43a5402ce75a
golang.org/x/sys v0.0.0-20200523222454-059865788121
google.golang.org/protobuf v1.24.0
gopkg.in/ini.v1 v1.56.0
```

Code Listings Examples

1. CreateBusinessPassport GraphQL mutation (Passport Data Service)

```
module Mutations
class CreateBusinessPassport < Mutations::BaseMutation
  description 'Creates a business passport'

  argument :full_legal_name, String, required: false
  argument :company_registration_number, String, required: false
  argument :legal_entity_name, String, required: false
  argument :business_pitch, String, required: false
  argument :website, String, required: false
```

```

    argument :number_of_employees, String, required: false
    argument :logo_id, String, required: false
    argument :presence_country_codes, [String], required: false do
      description "Dictionary name: countries"
    end
    argument :total_assets, String, required: false
    argument :total_liabilities, String, required: false
    argument :yearly_turnover_percentage, String, required: false
    argument :source_of_funds_code, String, required: false do
      description "Dictionary name: sources_of_funds"
    end
    argument :industry_codes, [String], required: false do
      description "Dictionary name: industries"
    end

    field :data, Types::BusinessPassport, null: true

    def resolve(**args)
      op =
        Commands::BusinessPassports::Create.(business_passport_params:
          args, current_account: context[:current_account])
      if op.success?
        {
          success: true,
          errors: [],
          data: op[:model]
        }
      else
        {
          success: false,
          errors: op[:errors] || []
        }
      end
    end
  end
end
end

```

2. CreateBusinessPassport Command (Passport Data Service)

```

module Commands
  module BusinessPassports
    class Create < ::Commands::Base
      include Import[:event_publisher]
    end
  end
end

```

```

    step :model
    step :save
    step :create_document_sections
    pass :send_notification
    step :link_to_current_account

  private

    def model(ctx, business_passport_params:, **)
      ctx[:model] =
        BusinessPassport.new(business_passport_params)
      ctx[:model].account_id = ctx[:current_account].id
      ctx[:model].region_code =
        ctx[:current_account].region_code
    end

    def save(ctx, model:, **)
      if model.save
        true
      else
        ctx[:errors] = model.errors
        false
      end
    end

    def create_document_sections(ctx, model:, **)
      [
        {kind: :company_formation, can_add_link: false},
        {kind: :license, can_add_link: true},
        {kind: :business_plan, can_add_link: false}
      ].each do |ds_attrs|

        Commands::DocumentSections::Create.(document_section_params:
          ds_attrs.merge(business_passport_id: model.id))
      end
    end

    def send_notification(ctx, model:, **)
      event_publisher.business_passport_created(model.reload,
        account: ctx[:current_account]) unless ctx[:skip_callbacks]
    end

    def link_to_current_account(ctx, model:, current_account:,
      **)
      op = Commands::AccountsBusinessPassports::Create.(
        accounts_business_passport_params: {
          business_passport_id: model.id,

```

```

        account_id: current_account.id,
        business_passport_region_code: model.region_code,
        can_manage: true
      },
      current_account: current_account
    )
    if op.success?
      true
    else
      ctx[:errors] = op[:errors]
      false
    end
  end
end
end
end
end

```

3. VerificationFlows Controller (Core Service)

```

module ClientManagement
  class VerificationFlowsController < BaseController

    get '' do
      policy_decision(:index?)
      json success(
        Serializers::Client::VerificationFlow.render_as_json(scope),
        meta: pagy_metadata(@pagy)
      )
    end

    get '/:id' do
      policy_decision(:show?)
      json
        success(Serializers::Client::VerificationFlow.render_as_json(resource))
    end

    post '' do
      policy_decision(:create?)
      op = Commands::Client::VerificationFlows::Create.(
        verification_flow_params: create_params,
        client: current_client,

```

```

        created_by: current_account
    )
    json respond_with_op(op,
Serializers::Client::VerificationFlow)
end

    put('/:id' do
        policy_decision(:update?)
        op = Commands::Client::VerificationFlows::Update.(
            id: params.fetch(:id),
            verification_flow_params: update_params,
            client_id: current_client.id
        )
        json respond_with_op(op,
Serializers::Client::VerificationFlow)
    end

    private

    def policy_object
        @policy_object ||=
Policies::Client::FormPolicy.new(pundit_user)
    end

    def create_params
resource_params(Contracts::Client::VerificationFlows::Create)
    end

    def update_params
resource_params(Contracts::Client::VerificationFlows::Update)
    end

    def resource_class
        VerificationFlow
    end

    def filter_collection(chain)
        Queries::Common::VerificationFlows::Get.(chain, params)
    end

    def association_chain
        super.where(client_id: current_client.id)
    end
end

```

```
def many_inclusion
  [
    :verification_steps
  ]
end

end
end
```